SESSION 5 LECTURE NOTES

Architecting GeoAI Systems on the Cloud: Platform Selection, Telecom Applications, and Cost Optimisation

GeoAI Fusion Bootcamp 2026

Introduction

A data-fusion script that runs correctly on a single laptop does not automatically scale to a production system serving millions of users. Scalable, secure cloud infrastructure is the non-negotiable core of modern GeoAI, connecting spatial databases, machine-learning compute, and streaming Internet-of-Things sources to the dashboards and mobile applications that decision-makers actually use.

This session addresses the practical problem of choosing among competing cloud ecosystems while controlling cost. Amazon Web Services, Microsoft Azure, and Google Cloud each offer distinct strengths for geospatial workloads, and a team locked into the wrong ecosystem faces compound integration friction later. Cost, meanwhile, escalates quickly once a workload moves from a small pilot project to a continental-scale production pipeline.

Participants in this session examine two applied telecom and infrastructure use cases, learn to compare the three major cloud ecosystems against clear selection criteria, and study a set of cost-optimisation practices that keep a GeoAI system financially sustainable at scale.

Discussion

The Cloud as the Central Nervous System of GeoAI

A GeoAI stack functions much like a nervous system, in which sensory input, central processing, and outward action must all connect seamlessly. Streaming and Internet-of-Things sources, such as sensors, electric-vehicle chargers, and grid systems, act as the sensory input. The cloud platform itself, spanning compute services, artificial-intelligence and machine-learning tools, and a spatial data platform built on technologies such as PostgreSQL, PostGIS, and BigQuery, acts as the central processing layer. Web dashboards, analytics reports, and mobile applications act as the outward action, delivering data-driven decisions back to the people who need them. Spatial databases in particular rely on cloud-backed durability and high-throughput input and output operations to remain reliable at scale.

Defining a Cloud Workload: Telecom Network Coverage

A concrete telecom example clarifies how a cloud workload is structured in practice. Raw telemetry, base-station logs, and frequency data arrive continuously through Internet-of-Things streams. Geographic-information-system mapping then spatially aligns towers, terrain, and building footprints. The cloud compute phase applies heavy parallel processing, using services such as Amazon SageMaker or Azure Machine Learning, to run a predictive coverage algorithm across this aligned data. Coverage analysis renders high-

resolution heatmaps for network optimisation, and a decision-support dashboard, built for instance in Power BI, integrates these results for planning teams. Cloud scalability becomes non-negotiable at the compute phase, since this is where massive, multi-variable spatial datasets meet deep-learning algorithms.

Use Case One: Telecom Hazard Response and Network Resilience

Natural hazards disrupt telecom networks and delay rescue operations at exactly the moment when connectivity matters most. A GeoAI solution predicts the likely impact of an approaching hazard, prioritises which actions to take first, and optimises the recovery sequence afterwards. A population-coverage estimation formula supports this work directly, calculating covered population as the sum, across a set of areas, of each area's population density multiplied by a binary coverage factor equal to one if the area is covered and zero if it falls within a dead zone. Building a micro-population heat map from this formula shows planners exactly where people actually congregate, helping to close the digital divide during a crisis.

Use Case Two: Electric-Vehicle Charging Infrastructure Planning

Planning electric-vehicle charging infrastructure requires fusing several layers that are rarely combined in a single analysis. Road-network data supports accessibility modelling, identifying where drivers can conveniently reach a charging point. Substation data supports grid-capacity prediction, since a charging station cannot be sited where the local grid cannot supply it. Electric-vehicle registration data and local income-level data together feed an optimisation engine that recommends where demand and capacity intersect most favourably.

From Sensor to Dashboard: The Infrastructure Blueprint

A complete GeoAI system follows a consistent blueprint regardless of the specific use case. Data sources, including Internet-of-Things sensors, electric-vehicle chargers, grid systems, and geographic-information-system data, feed into a cloud platform built on application servers and a spatial data platform such as PostgreSQL with the PostGIS extension. This platform, in turn, delivers results through web dashboards and mobile applications, translating spatial intelligence into a scalable, secure, cloud-native architecture.

Strategic Platform Matrix: AWS, Azure, and Google

Amazon Web Services offers massive scale and open data access as its core strength, with Amazon Location Service and SageMaker as key spatial tools, heavy graphics-processing-unit clusters and data lakes as its compute paradigm, and production engineers handling petabyte-scale workloads as its primary user base. Microsoft Azure offers enterprise business-intelligence integration as its core strength, with Synapse Spatial Analytics and Azure Maps as key tools, a unified data warehouse as its compute paradigm, and organisations already committed to Microsoft tools as its primary user base. Google Colab and Google Earth Engine together offer rapid prototyping and academic research as their core strength, with Earth Engine's petabyte-scale satellite archive as the key tool, an interactive Python and Jupyter environment as the compute paradigm, and academic professionals and data scientists as the primary user base.

The Prototyping Engine: Google Earth Engine and Colab

Google Earth Engine grants instant access to petabytes of historical satellite imagery and geospatial datasets without requiring the user to manage underlying storage. Google Colab complements this with a zero-friction Python execution environment, enabling academic professionals and innovators to build machine-learning

models directly in a browser. Together, these two tools form the ideal hub for bootcamps, hackathons, and certified assignments, since participants can prototype and share work without provisioning any infrastructure themselves.

Production Scale: The AWS Geospatial Ecosystem

Amazon Web Services supports production-scale workloads through three complementary services. Amazon S3 combined with the Open Data programme hosts Landsat and Sentinel archives at no storage cost, lowering the barrier to massive imagery processing. Amazon SageMaker moves raw satellite data to deployed prediction models without requiring the user to manage custom graphics-processing-unit clusters. Amazon Location Service provides managed maps, routing, and geocoding application programming interfaces without the need to negotiate separately with external map providers.

Enterprise Integration: The Azure Spatial Ecosystem

Microsoft Azure integrates spatial analysis directly into an organisation's existing business systems. Spatial Analytics within Azure Synapse answers geospatial questions alongside sales, customer records, and operational metrics within a single query environment, removing the need to export data elsewhere. Azure Cognitive Services automates feature extraction from imagery without requiring a separately stitched toolchain. Azure Maps then provides native integration for rendering and routing directly into existing Microsoft productivity tools, such as Power BI.

Strategic Platform Selection Criteria

Choosing among these ecosystems depends on three practical questions. The first concerns the scale of data involved. A pilot project suits Colab or Earth Engine, while a petabyte-scale daily pipeline suits Amazon Web Services, and matching infrastructure to actual scale avoids costly over-provisioning. The second concerns the existing ecosystem an organisation already uses, since staying within a familiar ecosystem limits compound integration friction. The third concerns skill availability, since the most technically capable platform is not necessarily the right choice if the team lacks the expertise to operate it confidently.

GeoAI FinOps: Cloud Cost Optimisation

Financial discipline is as important to a GeoAI system as technical design. Spot and reserved instances offer substantial discounts over on-demand pricing for interruptible batch-processing jobs, such as a large nightly reprocessing task. Storage tiering moves archival imagery and completed outputs to cold storage tiers automatically, governed by data-age rules. Right-sizing compute provisions instances to match actual workload requirements rather than defaulting to an oversized configuration chosen out of caution.

Delivering Measurable Impact

A well-architected GeoAI system ultimately delivers impact across three connected dimensions. Economic impact includes lower downtime, optimised operations, and increased return on investment for telecom and infrastructure spending. Social impact includes improved public safety, faster disaster-communication restoration, and progress toward bridging the digital divide. Environmental impact includes smarter resource use, a lower carbon footprint, and climate-resilient digital infrastructure. Together, these outcomes support the bootcamp's broader goal of stronger networks, faster response, and safer communities across a connected and resilient ASEAN.

Dual Frameworks for Innovation

Two complementary canvases guide participants from problem to solution. The GeoAI Innovator Canvas focuses on developing the person, the innovator capable of designing a responsible GeoAI solution. The Problem-Data-GeoAI-Solution Canvas focuses on developing the product, the innovation itself. A transformation pipeline connects the two. A problem statement leads to spatial context, which leads to GeoAI intelligence, which leads to a technology stack, which leads to value and impact, and finally to a pitch. Stakeholders, data requirements, solution design, ethics and privacy, and an implementation roadmap all support this pipeline at each stage.

Conclusion

This session showed that architecting a GeoAI system on the cloud requires more than selecting a capable platform. It requires matching that platform to the scale of the workload, the existing organisational ecosystem, and the team's own skill base, while applying disciplined cost-optimisation practices such as spot instances, storage tiering, and right-sizing.

The telecom hazard-response and electric-vehicle charging examples demonstrate that cloud-based GeoAI systems deliver economic, social, and environmental impact simultaneously. Participants carrying this session's frameworks forward into their own projects should be able to select an appropriate cloud ecosystem, justify that choice against the selection criteria introduced here, and estimate the cost implications of their design.